The Twin Paradox Revisited and Relativistic Velocity Addition

PHYS 2203 — Introductory Physics III

October 2, 2026

Overview

We start by applying the **relativistic Doppler effect** from Lecture 15 to the twin paradox, by having each twin transmit a pulse once a year and simply counting what arrives. We then derive **relativistic velocity addition**, the rule that replaces the Galilean $u' = u - v$ of Lecture 13, and check that it keeps light at speed c in every frame, reduces to the Galilean rule at low speeds, and never lets two subluminal speeds combine to exceed c.

1 Recap: The Relativistic Doppler Effect

A source at rest in frame O emits light of frequency f. An observer moving at speed v relative to the source measures a different frequency f'. Last lecture we derived it in two steps — a classical catch-up in the source frame, giving the interval between arrivals as measured in O, followed by a conversion to the observer's own clock, which is where the single factor of γ enters. The results were

$$\boxed{f' = f\sqrt{\frac{1 - v/c}{1 + v/c}}} \quad \text{(receding, red shift)}, \qquad \boxed{f' = f\sqrt{\frac{1 + v/c}{1 - v/c}}} \quad \text{(approaching, blue shift)}. \tag{1}$$

2 The Twin Paradox by Light Signals

Return to Anna and Bob of Lecture 15. Anna flies to a planet 6 light-years away at $v = 0.6c$, turns around, and comes home. Bob ages 20 years, Anna 16. We got that from lines of simultaneity. Here is the same answer derved a second way, using the Doppler effect.

Let each twin transmit a pulse once a year by their own clock, and let each simply count the pulses received. Pulses sent once per year arrive at the rate f' given by the two boxed formulas of Section 1, and at $v/c = 0.6$ those give

$$\frac{f'}{f} = \sqrt{\frac{1 - 0.6}{1 + 0.6}} = \sqrt{\tfrac{1}{4}} = \frac{1}{2} \quad \text{(receding)}, \qquad \frac{f'}{f} = \sqrt{\frac{1 + 0.6}{1 - 0.6}} = \sqrt{4} = 2 \quad \text{(approaching)}, \tag{2}$$

so a receding clock is *seen* to tick at half rate and an approaching one at double rate.

What Anna sees. Her trip lasts 16 years by her own clock, 8 out and 8 back. Her recession turns into approach at her own turnaround point, exactly halfway along her journey.

leg	duration (her clock)	factor	Bob's pulses received
outbound	8 yr	1/2	4
inbound	8 yr	2	16
total	16 yr		**20**

She counts 20 pulses. Bob aged 20 years.

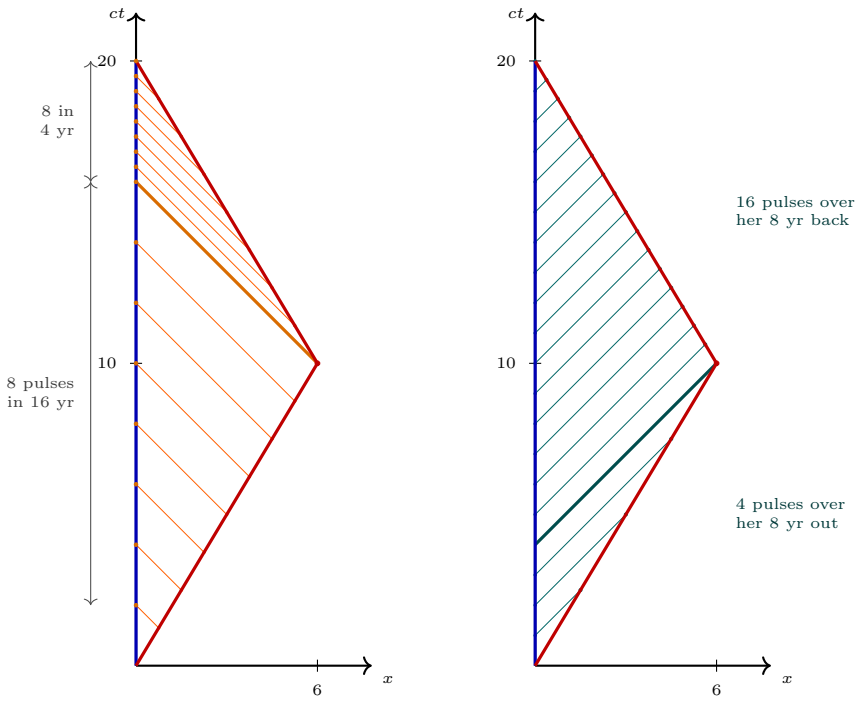
What Bob sees. Bob does *not* switch at his halfway point, because the news of the turnaround has to travel back to him. Anna turns around at $t = 10$ yr at a distance of 6 light-years, so that light reaches Bob at

$$t = 10 + 6 = 16 \text{ yr}, \tag{3}$$

with only 4 years of the trip left to run.

phase	duration (his clock)	factor	Anna's pulses received
red-shifted	16 yr	1/2	8
blue-shifted	4 yr	2	8
total	20 yr		**16**

He counts 16 pulses. Anna aged 16 years. Both twins agree, and neither ever had to decide what "now" meant far away.



(a) Anna's pulses reaching Bob

(b) Bob's pulses reaching Anna

Figure 1: Each twin transmits one pulse per year by their own clock. **(a)** Anna's pulses run back to Bob. The eight outbound ones reach him spread over 16 years, the eight inbound ones bunched into the final 4. The changeover is set by the heavy ray — the pulse sent *at* the turnaround, 6 light-years out, which does not arrive until $t = 10 + 6 = 16$ yr. **(b)** Bob's pulses run out to Anna. Only four reach her during the whole outbound leg, and the remaining sixteen during the inbound leg of equal length. Her changeover is set by the heavy ray, Bob's pulse of year 4, which arrives exactly *at* the turnaround event — the same event panel (a)'s heavy ray departs from. Bob counts 16 pulses in 20 years, Anna counts 20 in 16.

Where the asymmetry lives. The two columns of factors are identical. What differs is *when* each twin's shift changes over:

- Anna switches at the midpoint of her trip, 8 of 16 years — 50% of the way through.
- Bob switches at 16 of 20 years — 80% of the way through.

3 Relativistic Velocity Addition

In Lecture 13 the Galilean rule for combining velocities, $u' = u - v$, was shown to be incompatible with the second postulate: applied to light it gives $c - v$ rather than c. The Lorentz transformation must therefore give a different rule. As always, v is the relative speed of the two frames and u is the velocity of an *object* measured within a frame.

To find how velocities transform, take differentials of the Lorentz transformation:

$$\begin{aligned} x' = \gamma\,(x - vt) \quad &\Longrightarrow \quad \mathrm{d}x' = \gamma\,(\mathrm{d}x - v\,\mathrm{d}t), \\ y' = y \quad &\Longrightarrow \quad \mathrm{d}y' = \mathrm{d}y, \\ z' = z \quad &\Longrightarrow \quad \mathrm{d}z' = \mathrm{d}z, \\ t' = \gamma\Big(t - \frac{v}{c^2}x\Big) \quad &\Longrightarrow \quad \mathrm{d}t' = \gamma\Big(\mathrm{d}t - \frac{v}{c^2}\,\mathrm{d}x\Big)\,. \end{aligned} \tag{4}$$

x-component. Dividing $\mathrm{d}x'$ by $\mathrm{d}t'$, then dividing numerator and denominator by $\mathrm{d}t$,

$$\frac{\mathrm{d}x'}{\mathrm{d}t'} = \frac{\gamma\,(\mathrm{d}x - v\,\mathrm{d}t)}{\gamma\big(\mathrm{d}t - \frac{v}{c^2}\mathrm{d}x\big)} = \frac{\dfrac{\mathrm{d}x}{\mathrm{d}t} - v}{1 - \dfrac{v}{c^2}\dfrac{\mathrm{d}x}{\mathrm{d}t}} \qquad \Longrightarrow \qquad \boxed{u'_x = \frac{u_x - v}{1 - \dfrac{v\,u_x}{c^2}}.} \tag{5}$$

y- and z-components. The transverse coordinates are unchanged, but the time interval is not, so these components transform too:

$$\frac{\mathrm{d}y'}{\mathrm{d}t'} = \frac{\mathrm{d}y}{\gamma\big(\mathrm{d}t - \frac{v}{c^2}\mathrm{d}x\big)} \qquad \Longrightarrow \qquad \boxed{u'_y = \frac{u_y}{\gamma\left(1 - \dfrac{v\,u_x}{c^2}\right)}, \qquad u'_z = \frac{u_z}{\gamma\left(1 - \dfrac{v\,u_x}{c^2}\right)}.} \tag{6}$$

Classical limit. If $v \ll c$ and $u_x \ll c$, the denominator tends to 1 and $\gamma \to 1$, so

$$u'_x \approx u_x - v, \qquad u'_y \approx u_y, \qquad u'_z \approx u_z, \tag{7}$$

recovering the Galilean rule of Lecture 13.

Light travels at c in every frame. Test the rule on a light signal, $u_x = c$:

$$u'_x = \frac{c - v}{1 - \dfrac{vc}{c^2}} = \frac{c - v}{1 - v/c} = \frac{c\,(c - v)}{c - v} = c. \tag{8}$$

The second postulate is built in. This is exactly what the Galilean rule failed to do.

Speed limit. More generally, combining any two speeds less than c always yields a result less than c. No sequence of boosts can push a particle past the light barrier.

Worked example: two rockets leaving a space station

Two rockets leave a space station. In the station frame, Rocket 1 moves in the $+x$ direction at $0.6c$, while Rocket 2 moves in the $+y$ direction at $0.8c$. What is the speed and direction of Rocket 2 as measured by Rocket 1?

Work in the frame of Rocket 1, so $v = 0.6c$ and $\gamma = 1.25$, and for Rocket 2 in the station frame $u_x = 0$, $u_y = 0.8c$:

$$u'_x = \frac{u_x - v}{1 - \dfrac{v\,u_x}{c^2}} = \frac{0 - 0.6c}{1 - 0} = -0.60c \qquad \text{(same as the classical value)}, \tag{9}$$

$$u'_y = \frac{u_y}{\gamma\left(1 - \dfrac{v\,u_x}{c^2}\right)} = \frac{0.8c}{1.25\,(1 - 0)} = 0.64c \qquad \text{(classically it would be } 0.8c\text{)}. \tag{10}$$

The resultant speed is

$$u' = \sqrt{u_x'^2 + u_y'^2} = \left[0.60^2 + 0.64^2\right]^{1/2} c = \boxed{0.88c}\,. \tag{11}$$

directed into the second quadrant at an angle $\theta' = \tan^{-1}(0.60/0.64) = 43^\circ$ from the $+y'$ axis.

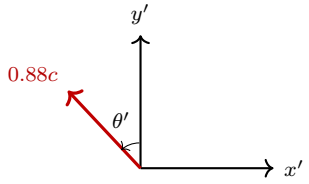


Figure 2: Velocity of Rocket 2 as measured in the Rocket 1 frame: $0.88c$, directed 43° from the $+y'$ axis toward $-x'$.

Comparison with the classical result. Galilean addition would give $u_x' = -0.60c$ and $u_y' = 0.8c$, hence

$$u' = \sqrt{0.60^2 + 0.80^2}\, c = c.$$

The classical prescription produces a rocket moving at exactly the speed of light. The relativistic result correctly stays below c.

4 Momentum Conservation

The most useful law we have is conservation of momentum, $\mathbf{P}_{\text{total}} = \sum_i m_i \mathbf{u}_i = \text{constant}$ for an isolated system. The first postulate demands that such a law hold in *every* inertial frame, not just in one. That is a real constraint, and it is worth checking before assuming the law carries over.

Classically the check passes. Using the Galilean rule $u_i' = u_i - v$ of Lecture 13,

$$P'_{\text{total}} = \sum_i m_i(u_i - v) = \sum_i m_i u_i - v\sum_i m_i \qquad \Longrightarrow \qquad P'_{\text{total}} = P_{\text{total}} - v\sum_i m_i. \tag{12}$$

The two totals differ by a *constant*, since the system's mass does not change and v is fixed. So if P_{total} is unchanged by a collision, so is P'_{total}. That is exactly what the first postulate demands of a law of physics.

But the Galilean rule is wrong. Redo it with the relativistic rule of Section 3. Two particles of mass 4 and 3 (arbitrary units) collide head-on in frame O, as drawn. Their classical momenta are equal and opposite before *and* after, so $\sum_i m_i u_i = 0$ throughout and classical momentum is conserved in this frame.

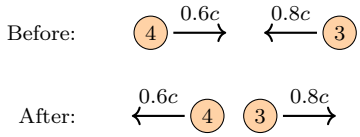


Figure 3: A collision constructed so that the *classical* total momentum vanishes in frame O, both before and after.

Transform all four velocities into a frame O' moving at $v = 0.6c$, using $u' = \dfrac{u - v}{1 - vu/c^2}$:

$$\text{before, mass 4:} \quad u' = \frac{0.6c - 0.6c}{1 - (0.6)(0.6)} = 0, \tag{13}$$

$$\text{before, mass 3:} \quad u' = \frac{-0.8c - 0.6c}{1 - (-0.8)(0.6)} = \frac{-1.4c}{1.48} = -0.946c, \tag{14}$$

$$\text{after, mass 4:} \quad u' = \frac{-0.6c - 0.6c}{1 - (-0.6)(0.6)} = \frac{-1.2c}{1.36} = -0.882c, \tag{15}$$

$$\text{after, mass 3:} \quad u' = \frac{0.8c - 0.6c}{1 - (0.8)(0.6)} = \frac{0.2c}{0.52} = 0.385c. \tag{16}$$

Now add up the classical momenta in the new frame:

$$\text{before:} \quad \sum_i m_i u_i' = 4(0) + 3(-0.946c) = -2.84c, \tag{17}$$

$$\text{after:} \quad \sum_i m_i u_i' = 4(-0.882c) + 3(0.385c) = -2.38c. \tag{18}$$

The problem. $\sum_i m_i u_i$ is conserved in frame O but *not* in frame O'. A conservation law that holds in one inertial frame and fails in another violates the first postulate, so classical momentum $m\mathbf{u}$ cannot be correct.

So $\mathbf{p} = m\mathbf{u}$ must be replaced. Notice what the argument does *not* tell us: it says the classical expression is wrong, but not what the right one is. We will derive it next lecture.